

Tribhuvan University
Institute of Science and Technology
 4 Years Bachelor in Computer Science and Information Technology (B.Sc.CSIT)
Entrance Examination
2071

Full Marks: 100
Time: 2.00 hrs.

Attempt all question:

Mathematics

(25 × 1 = 25)

1. If $(x - 3) = 4$ then x is
 (a) 7 and -1 (b) 7 and 1 (c) 7 (d) 1
2. If $\{x - x^2 = 9 \text{ and } 2x = 6\}$
 (a) ϕ (b) $\{3\}$ (c) $\{3, -3\}$ (d) $\{4, 3\}$
3. If $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 + 3$ then $f^{-1}(4)$ is
 (a) ϕ (b) $\{-1, 1\}$ (c) $\{1, 0\}$ (d) $\{4, 3\}$
4. The value of $\log_2 \sqrt{2} + \log_3 \sqrt[3]{3} + \log_4 \sqrt[4]{4}$ is
 (a) 0 (b) $\frac{13}{12}$ (c) $-\frac{13}{2}$ (d) $\frac{\pi}{6}$

5. If $\cos^2 \theta = \frac{1}{2}$, then θ is

- (a) $n\pi \pm \frac{\pi}{4}$ (b) $n\pi + \frac{\pi}{4}$ (c) $n\pi \pm \frac{\pi}{3}$ (d) $2n\pi$

6. $\cos(\sin^{-1} 3/5) =$

- (a) $3/5$ (b) $4/5$ (c) $9/25$ (d) $4/25$

7. If a^2, b^2, c^2 are in H.P. then

- (a) $2a^2c^2 = b^2(a^2 + c^2)$ (b) $a^2c^2 = b^2(a^2 + c^2)$
 (c) $2b^2c^2 = a^2(b^2 + c^2)$ (d) $c^2a^2 = 2b^2(a^2 + c^2)$

8. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $A^2 - nA - I = 0$, then the value of n is

- (a) 0 (b) 4 (c) 1 (d) -4

9. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, then A is

- (a) Symmetric (b) Skew-symmetric
 (c) Singular (d) Non singular

10. The system of equations $2x + y = 5$; $4x + 2y = 8$ is

- (a) consistent and dependent
- (b) inconsistent and independent
- (c) inconsistent and dependent
- (d) consistent and independent

11. The value of $i^7 + 1/i^4$ is

- (a) 1
- (b) -1
- (c) -i
- (d) -i + 1

12. The value of $\frac{x - wy + w^2z}{y + wz + w^2x}$, w is the cube root of unity, is

- (a) w
- (b) $x + y + z$
- (c) w^2
- (d) 1

13. If the roots of $x^2 + 4x + k = 0$, are equal then k is

- (a) 0
- (b) 4
- (c) -4
- (d) 16

14. The two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are identical if

- (a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- (b) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
- (c) $\frac{a_1}{a_2} = \frac{b_1}{b_2}$
- (d) $\frac{b_1}{b_2} = \frac{c_1}{c_2}$

15. The two lines $2x^2 - 2xy + ky^2 = 0$ are perpendicular to each other if k is

- (a) 2
- (b) -2
- (c) 0
- (d) 1

16. The equation of the circle touching the positive x - axis and y - axis with radius 2 units is

- (a) $x^2 + y^2 - 4x - 4y + 4 = 0$
- (b) $x^2 + y^2 + 4x + 4y + 4 = 0$
- (c) $x^2 + y^2 - 4x + 4y + 4 = 0$
- (d) $x^2 + y^2 - 4x - 4y - 4 = 0$

17. If the line $4x + 3y = k$ is tangent to the circle $x^2 + y^2 = 2$ at $(2, 1)$ then k is

- (a) 25
- (b) -25
- (c) ± 25
- (d) ± 11

18. The value of $\lim_{x \rightarrow 2} \frac{x^5 - 2^5}{x - 2}$ is

- (a) 80
- (b) 40
- (c) 10
- (d) 0

19. The value of $\lim_{x \rightarrow 0} \frac{3x + 4 \tan 2x}{x}$ is

- (a) 7
- (b) 3
- (c) 11
- (d) 4

20. The derivative of $y = b^x$ is

- (a) $x b^{x-1}$ (b) b^x (c) $x^b \ln b$ (d) $b^x \ln b$

21. The derivative of $y = \cos(\cos \sqrt{x})$ is

- (a) $\frac{\sin(\cos \sqrt{x}) \sin \sqrt{x}}{2\sqrt{x}}$ (b) $\sin(\cos \sqrt{x})$
(c) $\sin(\sin \sqrt{x})$ (d) $\frac{\sin(\cos \sqrt{x})}{2\sqrt{x}}$

22. The value of x for which the derivative of $x^2 + 2x + 2$ is zero is

- (a) 1 (b) 2 (c) 0 (d) -1

23. The point of inflection of the curve $x^3 + 3x^2 - 9x + 2$ is

- (a) -1 (b) 1 (c) 6 (d) 0

24. $\int \frac{dx}{1 - \cos x} =$

- (a) $\tan x/2$ (b) $-\cot x/2$
(c) $\cos x/2$ (d) $-\tan x/2$

25. $\int_0^{\pi/2} \frac{\cos x}{1 + \sin x} dx =$

- (a) ∞ (b) $-\ln 2$ (c) 0 (d) 1

English

(25×1=25)

26. The word “education” has the stress on the syllable.

- (a) first (b) third
(c) second (d) fourth

27. Select the right synonym to the word “arduous”.

- (a) simple (b) difficult
(c) easy (d) strange

28. Give the opposite meaning of “obscure”.

- (a) pedantic (b) implicit
(c) explicit (d) obnoxious

29. The initial consonant sound of “chef” is the same as that of

.....

- (a) chute (b) chess
(c) child (d) cheap

30. Which of the following pair has the same vowel sound?

- (a) hut/hot (b) sit/seat
(c) salt/mark (d) pair/tear

31. I completely differ you this point.

- (a) from; at (b) from; in
(c) from; on (d) to; at

32. We feel sympathy the poor

- (a) to (b) with (c) about (d) for

33. He insister leaving immediately.

- (a) at (b) in (c) on (d) over

34. Tick the correct word to the underlined expression.

No one knows his hole and corner method of dealing with people.

- (a) strict (b) secret
(c) servile (d) suspicious

35. Food that agrees to one's taste is

- (a) pungent (b) edible
(c) palatable (d) sensuous

36. The word "pleased" takes the prefix

- (a) un - (b) mis -
(c) dis - (d) none -

37. One who is present every where is

- (a) omnipresent (b) omniscient
(c) theist (d) omnipotent

38. Fear of food is called as

- (a) sitophobia (b) ergophobia
(c) panthophobia (d) hycophobia

39. A funeral bell is called.....

- (a) call (b) knell
(c) replica (d) invitation

40. People a large number of computers now, because prices are getting low.

- (a) must buy
(b) must have bought
(c) must be buying
(d) must buying

41. Although they are similar, chipmunks are most other ground squirrels.

- (a) alike (b) like as
(c) like to (d) like

42. She the whole day long.

- (a) has been working (b) works
(c) had worked (d) worked

43. Now a days Shyam is busy to lake care of his health.

- (a) very (b) too
(c) so (d) extremely

44. She demanded her status

- (a) know (b) knowing
(c) to know (d) to be known

45. The mail comes at ten O'clock everyday saturday.

- (a) but also (b) and
(c) also (d) but

46. In my office, salary is paid by month.

- (a) the (b) a
(c) an (d) some

47. Every boy and every girl very responsible

- (a) are (b) is
(c) has (d) have

48. Which of the following is accepted?

- (a) some of rice is still left.
(b) time and tide wait for none.
(c) she absents herself from the college today.
(d) None of the squares was properly drawn.

49. Which of the following is a complex sentence?

- (a) However, hard she worked, she failed in the exams.
(b) It rained heavily, however he came to school.
(c) In spite of the heavy rain, he come to school.
(d) Neither she nor he will come.

50. Which of the following is an adjective?

- (a) sympathize (b) Lighten
(c) perception (d) moral

Physics

(25×1=25)

51. The dimensions of (h/e) are same as that of

- (a) magnetic field (b) electric field
(c) magnetic flux (d) electric flux

52. The number of significant figures in 0.0030×10^{11} m are
(a) 5 (b) 4 (c) 11 (d) 2
53. μ_s and μ_k are coefficients of static and kinetic frictions between two bodies. Then
(a) $\mu_s \geq \mu_k$ (b) $\mu_s < \mu_k$
(c) $\mu_s = 0$ and $\mu_k > 0$ (d) $\mu_s < 0$ and $\mu_k < 0$
54. When a torque ($\vec{\tau}$) acts on a body, the change in angular momentum ($\vec{L}$) of the body is
(a) perpendicular to $\vec{L}$ (b) parallel to $\vec{L}$
(c) always zero (d) 180° to $\vec{L}$
55. The Poisson's ratio of a material can't be
(a) 0.10 (b) 0.20 (c) 0.40 (d) 0.65
56. The depth of water at which air-bubble of radius 0.4 mm may remain in equilibrium (surface tension of water = 72×10^{-3} N/m, $g = 9.8$ m/s²) is given by
(a) 7.348 cm (b) 0.918 cm
(c) 3.674 cm (d) 1.837 cm
57. When a cylinder is heated, its length increases by 2% , the area of its base will increase by :
(a) 0.5% (b) 1% (c) 2% (d) 4%

58. In a pressure cooker food is cooked usually at a pressure
(a) one atmosphere (b) half atmosphere
(c) two atmosphere (d) 50 atmosphere
59. The mean free path of a gas varies with absolute temperature (T) as
(a) T^4 (b) $1/T^2$ (c) $1/T$ (d) T
60. The efficiency of cannot engine working between steam point and ice point is:
(a) 16.8% (b) 26.81% (c) 36.8% (d) 46.8%
61. The bending of sound waves through the corners of the obstacle is called
(a) refraction (b) Interference
(c) beats (d) diffraction
62. When a tuning fork is vibrated its prongs have a phase difference of:
(a) 0 (b) $\pi/4$
(c) π (d) 2π

63. The image of an object placed at the focus of convex mirror is at a distance
(a) ∞ (b) f (c) $2f$ (d) $\frac{f}{2}$
64. If sky is seen from moon's surface, it will appear
(a) black (b) blue (c) white (d) red
65. A double convex air bubble in a glass sheet would behave as a
(a) convergent lens
(b) concave mirror
(c) divergent lens
(d) plane mirror
66. When two prisms are combined it gives
(a) reflection only
(b) lateral shift
(c) dispersion and deviation both
(d) dispersion or deviation
67. Which property of electrons is used in an electron microscope
(a) negative charge (b) wave nature
(c) spin (d) small size
68. A system of three charges may be in equilibrium if they are placed
(a) on a straight line
(b) on corners of equilateral triangle
(c) on three corners of a rectangle
(d) On the corners of a cube
69. The radius of Earth is 6400 km. The capacitance is
(a) 6400 MF (b) 6400 F
(c) 1 F (d) 711 μF
70. Two wires of same material have lengths L and $2L$ and cross section areas $4A$ and A respectively. The ratio of their specific resistances would be
(a) 1 : 1 (b) 1 : 8 (c) 8 : 1 (d) 1 : 2
71. The magnetic field inside a long solenoid is
(a) zero (b) $\mu_0 n i$ (c) $n i$ (d) $\mu n i$
72. Electrons in the atom are held due to
(a) Nuclear forces (b) Coulomb forces
(c) Van der Waals force (d) Gravitational forces

Name.....

Roll No.....

4 Years B.Sc.CSIT Ent. 2071

73. Which series of H atom lies in visible region?

- (a) Lyman (b) Bracket and Paschen
(c) Balmer (d) Paschen

74. The half life of a substance is 1600 years. The mean life is

- (a) 1600 years (b) 3200 years
(c) 800 years (d) 2309 years

75. Rydberg constant is

- (a) Same for all elements
(b) different for different elements
(c) a universal constant
(d) is different for lighter elements but same for heavier elements.

Chemistry

(25×1=25)

76. How many orbitals are possible for principal quantum number 3?

- (a) 3 (b) 4 (c) 6 (d) 9

77. The number of moles of water present in 18g of water is

- (a) 18 (b) 1 (c) 1.8 (d) 180

78. The half life of an radioisotope is 30 minutes. After how much time the amount of radioisotope left is $\frac{1}{4}$ th of the original amount.

- (a) 4 minutes (b) 30 minutes
(c) 60 minutes (d) 120 minutes

79. What is oxidation of chlorine is HOCl

- (a) + 1 (b) 0 (c) - 1 (d) + 3

80. Haemoglobin contains 0.33% of iron by weight. Molecular weight of haemoglobin is 67200. The number of iron atoms (At wt of Fe = 55.85) in one molecule of haemoglobin is

- (a) 2 (b) 4 (c) 6 (d) 8

81. The gas law which deals with the effect of pressure in a given volume of gas at constant temperature is known as

- (a) Boyle's Law (b) Charle's Law
(c) Graham's Law (d) Dalton's Law

82. Electronic configuration of four elements A, B, C, D are as follow, which will be most metallic?

- (a) A = 2, 8, 4 (b) 2, 8, 8
(c) 2, 8, 1 (d) 2, 8, 7

Name.....

Roll No.....

4 Years B.Sc.CSIT Ent. 2071

83. In which of the following compound the molecular geometry can not be predicted by the use of VSEPR theory

- (a) NaCl (b) H₂O (c) CH₄ (d) NH₃

84. Consider the reaction $2A + B \rightarrow C + D$. If the concentration of the reactants are increased by three times, the rate of reaction will increase by

- (a) 3 times (b) 6 times
(c) 9 times (d) 27 times

85. An aqueous solution of NaCl would be

- (a) acidic
(b) basic
(c) neutral
(d) Could be acidic or basic depending upon temperature.

86. The pH of pure water at 50°C would be

- (a) 7.0 (b) more than 7.0
(c) less than 7.0 (d) can not be predicted

87. Which of the following metals generate H₂ on treatment with dil HNO₃

- (a) Zn (b) Mn
(c) Cu (d) Pb

88. 1 M H₂ SO₄ is equal to

- (a) 1 N H₂ SO₄ (b) 2 N H₂ SO₄
(c) 0.5 N H₂ SO₄ (d) 4 N H₂ SO₄

89. Hydrogen molecule which has parallel spin of two nuclei is called

- (a) ortho hydrogen (b) para hydrogen
(c) heavy hydrogen (d) nascent hydrogen

90. Ammonia is manufactured by the use of

- (a) Solvay's Process (b) Haber's Process
(c) Ostwald's Process (d) Down's Process

91. Haematite is an ore of

- (a) Fe (b) Cu (c) Zn (d) Hg

92. Which of the following is not a nucleophile

- (a) BF₃ (b) NH₃
(c) H₂O (d) OH⁻

93. Anti-Markovnikoff addition of HBr is not observed in

- (a) Propene (b) 1-butene
(c) 2-butene (d) Pent-2-one

Name.....

Roll No.....

4 Years B.Sc.CSIT Ent. 2071

94. When Oxalic acid crystals and glycerol are mixed and heated to about 110°C , the compound obtained is

- (a) chloroform
- (b) diethylether
- (c) acetone
- (d) formic acid

95. Which of the following is a meta directing group

- (a) $-\text{NH}_2$
- (b) $-\text{OH}$
- (c) $-\text{COOH}$
- (d) $-\text{R}$

96. How many position isomers are possible for dichlorobenzene

- (a) 2
- (b) 3
- (c) 4
- (d) 6

97. Which one of the following is a condensation polymer

- (a) Polyethene
- (b) Terylene
- (c) Pvc
- (d) Teflon

98. Pencillin is an example of

- (a) antiseptic drug
- (b) antipynetic drug
- (c) antibiotic drug
- (d) analgesic drug

99. Enzymes are

- (a) carbohydrates
- (b) proteins
- (c) lipids
- (d) nucleic acids

100. Which base is found only in the nucleotides of RNA?

- (a) adenine
- (b) guanine
- (c) Uracil
- (d) cytosine